
GY457 Applied Urban and Real Estate Economics

LSE GY457 - AT 2025
Introduction
Sara Bagagli

Autumn term lecturer

Sara Bagagli

- AT LSE since 1 year
- Grew up in Italy; Economics PhD in Zurich; Economics Postdoc at Harvard
- Research on urban economics and transportation
- What drives the unequal distribution of people and economic activity across space
 - The (Express)Way to Segregation: Evidence from Chicago
 - The Fastest Route to Specialization? Evidence from the Expansion of the Italian Highway System
 - No Way to the Highway: The Political Economy of Highway Revolts
- Office hours: To be booked via Student Hub

GY457 team

- Me for lectures and seminars in AT
- Prof. Christian Hilber (WT)
 - Course convenor
 - Responsible for admission
 - Responsible for exam



What we expect from you: Study

- Useful to go through core readings before class
- After class:
 - Review slides, notes
 - Make sure you understand the theories and the logic/intuition
 - Follow up on readings if necessary (also ask for additional readings if needed)
 - Can you connect theory to evidence?
 - Own thoughts on interesting applications
- Actively participate, with us and among yourselves
 - Ask and answer questions, also on Moodle! (Q&A Forums)
 - Interactive classes much better and fun + foster learning

What we expect from you: Seminars

- Seminars are compulsory: We have to record attendance
- Idea: Apply concepts in smaller groups (see schedule)
 - Consist of: Exercises to solve, paper presentations, case studies
- You lead the way!
 - Presenters prepare presentations: engage with audience, ask questions, could use interactive learning tools (e.g., menti)
 - Other students: Engage with presenters, ask and answer questions, challenge,...
- Specific tasks outlined in the seminar handout (All students must come prepared)

What we expect from you: Collaborate

- Work in groups
 - Prepare seminars, help each other, discuss readings
- Great cohort for networking
 - Collectively, you have a remarkable degree of experience, expert knowledge, and diversity
- Benefit from one another
- & Benefit from the environment you are at
 - Research-based teaching: Teaching staff experts in topics
 - LSE!! (research seminars, public lectures,...)

Teaching approach

- Modern Economics course
- Use models: Does a theory explain one aspect in world?
 - ... Using reasonable assumptions?
- Does evidence support the theory?
 - Are predictions in line with causal evidence?
 - Always keep in mind distinction between *correlation* and *causation*
- Apply model to do predictions and find best policy,...
- Learn (i) how to use a model and which assumptions are critical for its functioning (understand its limitations); (ii) how to do robust empirical evidence

Studying at LSE

- LSE motto “To know the causes of things”
 - With a combination of theory, evidence, and analytical skills
 - LSE is a world-class social science research university
 - Unique pool of highly respected research, knowledge, and expertise
- REEF/LSE degree has value because of signaling and selection
 - Challenging to engage with theory and empirics!
 - You have been selected because we think you will make it
 - REEF degree signals you are “special,” have analytical skill
- Commit to LSE agenda and strengths: Academic rigor

Expected learning outcomes

- Introduce urban economics and applications
- Show how theory and evidence help solving real-world problems
 - Make good economic decisions (maximize welfare)
 - Make good financial decisions (maximize profits)
- Seminars connect to concrete financial decisions
- Key aspect: What you learn can be useful in real life
 - One more note on this at the end of the lecture

Course structure

- AT
 - Block 1: Intro to agglomeration and regional economic forces
 - Block 2: City structure and spatial organization
 - Block 3: Urban transformation and social dynamics
- WT (with Christian)
 - Block 4: Interlinkages between government and housing markets
 - Block 5: Urban housing markets
 - Block 6: Commercial real estate

AT Structure

- Block 1

- Intro to Urban and Regional Economics and course overview
- Topic 1: Regional and urban concentration forces
- Topic 2: The empirics of agglomeration
- Topic 3: The costs and benefits of agglomeration

- Block 2

- Topic 4&5: Monocentric city model
- Topic 6: Firm location choice

- Block 3

- Topic 7: The vertical dimension of cities
- Topic 8: Neighborhoods and segregation
- Topic 9: Gentrification and suburbanization
- Topic 10: Measuring cities with new technologies

Logistics: Exam

- 3h (+10 *min reading time*) in-person exam during ST
 - Choice of questions with restrictions
 - Cover material from Autumn and Winter terms
 - Use of graphs useful and sometimes required
 - Math can be helpful but not required
- Answer to the point: No credit for content not asked
- Good answers combine: reference to theory, evidence, *and* own thoughts
- Possibly useful materials
 - Past exams
 - Revision session with details on exam in *Winter term*

Logistics: Reading list

- Reading list available online
 - Accessible through Moodle page of the course
- Two types of readings: core and complementary
- Core readings (compulsory):
 - Good to know in some detail
 - Get the big picture (no need to memorize single numbers)
 - Look for aspects covered in lectures/seminars
- Other readings:
 - May be useful to check what is interesting and get key message
- Lecture slides list all the references at the end
- Seminars may have their own compulsory readings

Logistics: Reading list

- Mix of textbook chapters and journal articles
- Journal articles should all be accessible via LSE library
 - Can also be accessed via publisher websites with LSE network
- Textbooks available online if there is ebook subscription
- Other textbooks available as hard copies at LSE library
- Some chapters have been scanned
 - Linked to Reading list @ LSE
 - But there are copyright restrictions

Logistics: Seminars

- Seminars are held in groups, week 3-4 & week 7-11
- Refer to seminar allocation document for
 - Allocation to seminar groups
 - Assignments of presentations
 - [COMING SOON](#) on Moodle
- Always prepare for the next seminar
 - Read seminar outline carefully
 - All students are asked to carry out homework
 - Presenters should prepare presentation and upload it on Moodle
- Seminars material is part of the exam (!)

Logistics: Feedback policy

- 3 hours (+10 *min reading time*) in person mock exam in week 8 Winter Term
 - Taking the mock exam is optional
 - Mimic real exam (2 AT + 2 WT questions, choose 1 AT + 1 WT)
 - Answer outline during revision session in week 11 WT

Logistics: Syllabus content

- Available on Moodle
- Administrative information
- Course overview (main topics)
- Expected schedule
- Notes on reading list
- Seminar information
- Advices on how to
 - a. Write essays
 - b. Prepare presentations

Logistics: Moodle

- Keep an eye on Moodle page
- Most relevant documents will be published there
 - Lecture handouts
 - Seminar allocation
 - Seminar outlines
 - Student presentations
 - Link to reading list
 - Syllabus
 - etc.

Outline

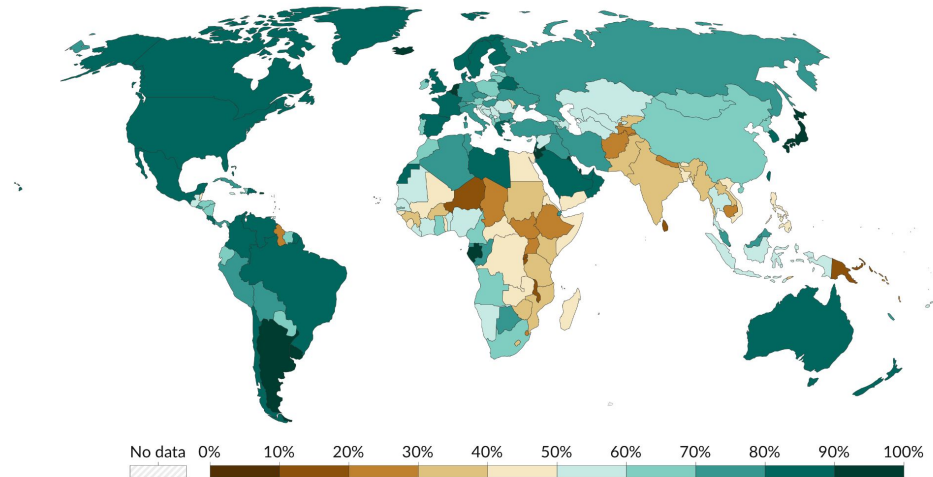
1. Teaching team and course logistics
2. **Motivation: Why urban economics?**
3. Case study: Tang (2024)

We live in an urbanized world

Share of the population living in urban areas, 2025

Our World
in Data

Share of the total population living in urban areas, given as estimates by HYDE until 2023 and UN projections through 2050, using the median fertility scenario for future urbanization trends.



Data source: United Nations, Department of Economic and Social Affairs, Population Division (2018); HYDE (2023)

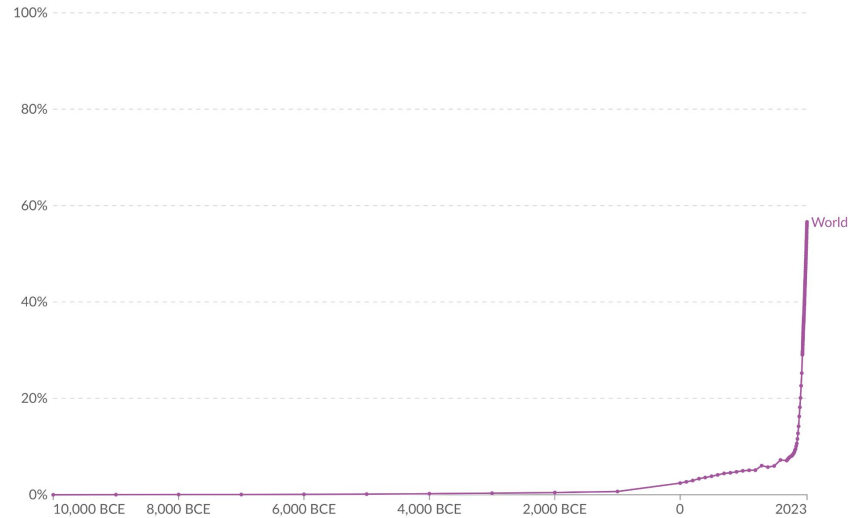
Note: Because the estimates of city and metropolitan areas are based on national definitions of what constitutes a city or metropolitan area, cross-country comparisons should be made with caution.

OurWorldinData.org/urbanization | CC BY

...which in the past 12,000 years got more and more urbanized

Share of the population living in urbanized areas, 10,000 BCE to 2023

Our World
in Data

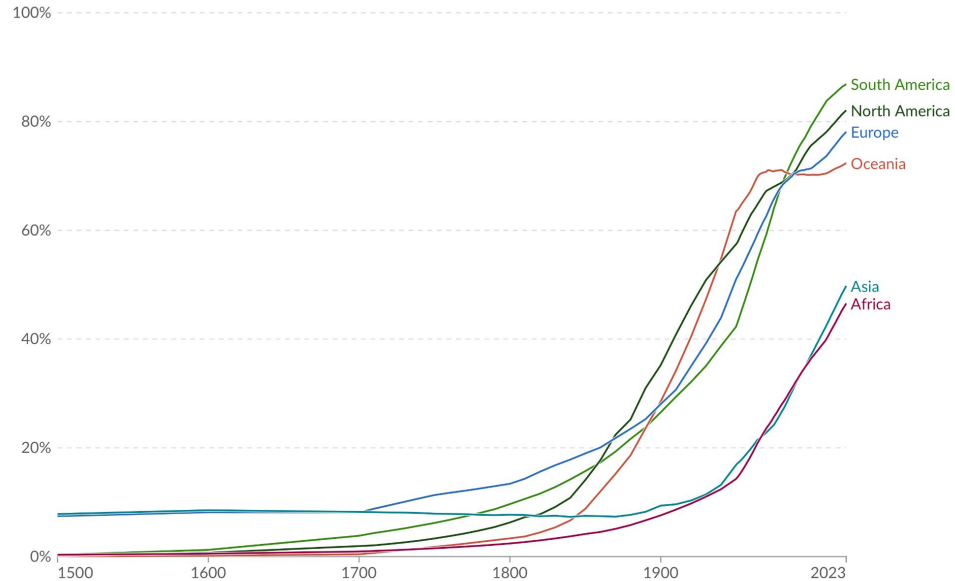


Data source: HYDE (2023)

OurWorldinData.org/urbanization | CC BY

...everywhere

Share of the population living in urbanized areas, 1500 to 2023



Data source: HYDE (2023)

OurWorldinData.org/urbanization | CC BY

Why considering geography: Most global economic inequality is *spatial*



Data source: Lakner and Milanovic (2015) – *Global Income Distribution: From the Fall of the Berlin Wall to the Great Recession*, World Bank Economic Review.

Other Asia refers to Asia without India, China, Hong Kong, Israel, Japan, Korea, Singapore, and Taiwan.

Developed countries are the EU-27, Australia, Bermuda, Canada, Hong Kong, Iceland, Israel, Japan, Korea, New Zealand, Norway, Singapore, Switzerland, Taiwan, and the United States.

The categorisation of countries is stable over the entire time period 1988-2011.

The data visualization is available at OurWorldinData.org. There you find more visualizations on this topic.

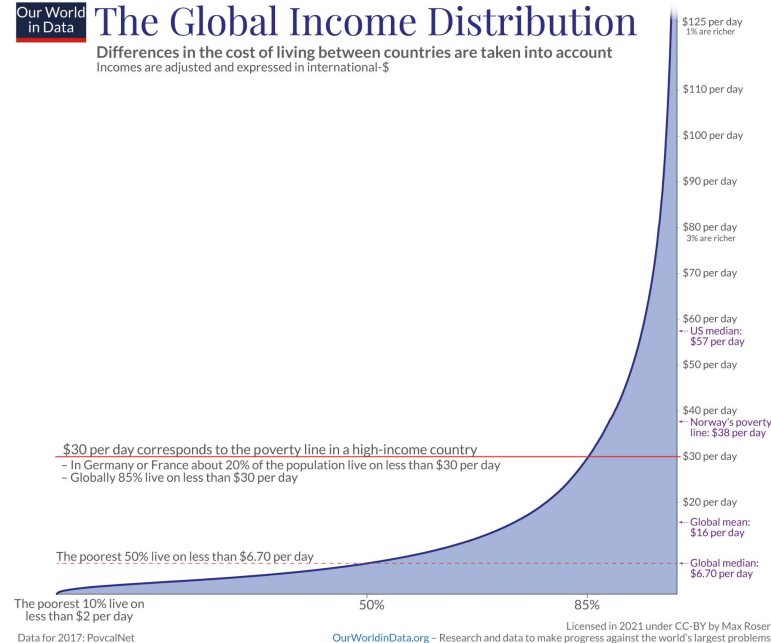
Licensed under CC-BY-SA by the authors Zdenek Hynek and Max Roser.

Not *who* you are, but *where* you are matters most for your living conditions

Our World
in Data

The Global Income Distribution

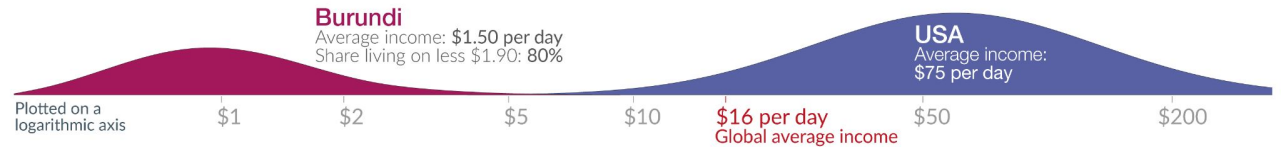
Differences in the cost of living between countries are taken into account
Incomes are adjusted and expressed in international-\$



Inequality *within* vs. *between* countries

Non-monetary sources of income are taken into account.

Expressed in *international-dollars*, which means that it is adjusted for price differences between the two countries.



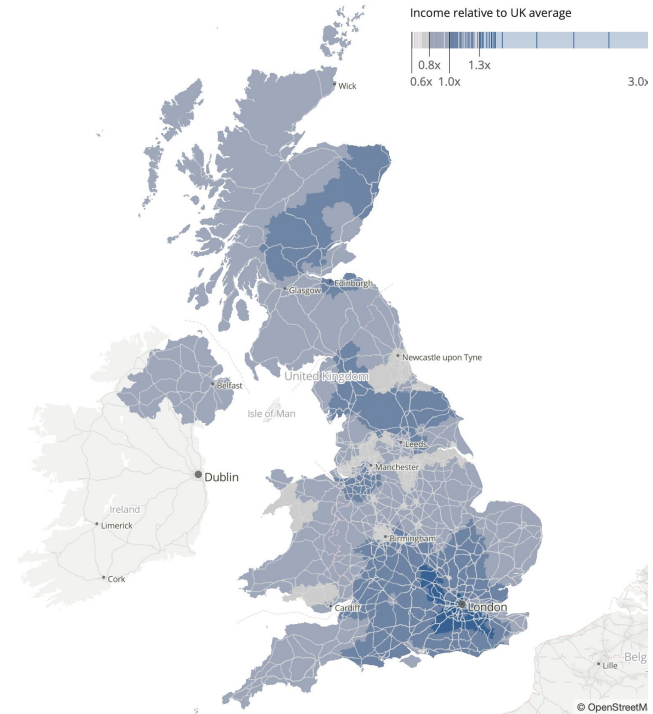
Data: PovcalNet for 2019 OurWorldinData.org – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the author Max Roser

- Extreme example: Burundi income distribution vs. US income distribution
- US highly unequal country
 - The two distributions do not overlap
 - Much of global inequality happens between countries

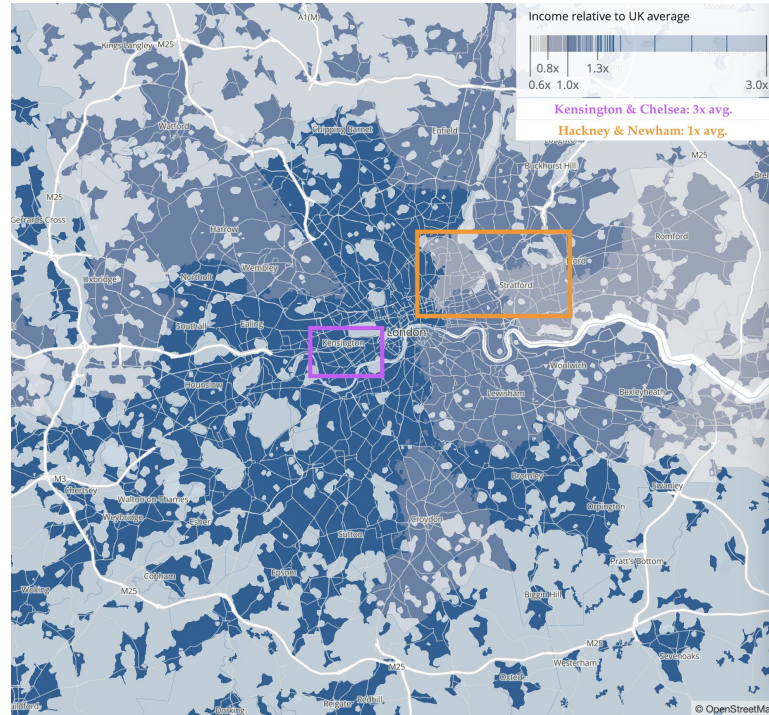
Within countries: UK

- Spatial distribution of household income in UK
 - Relative to UK average (£21,000)
 - Subregional level



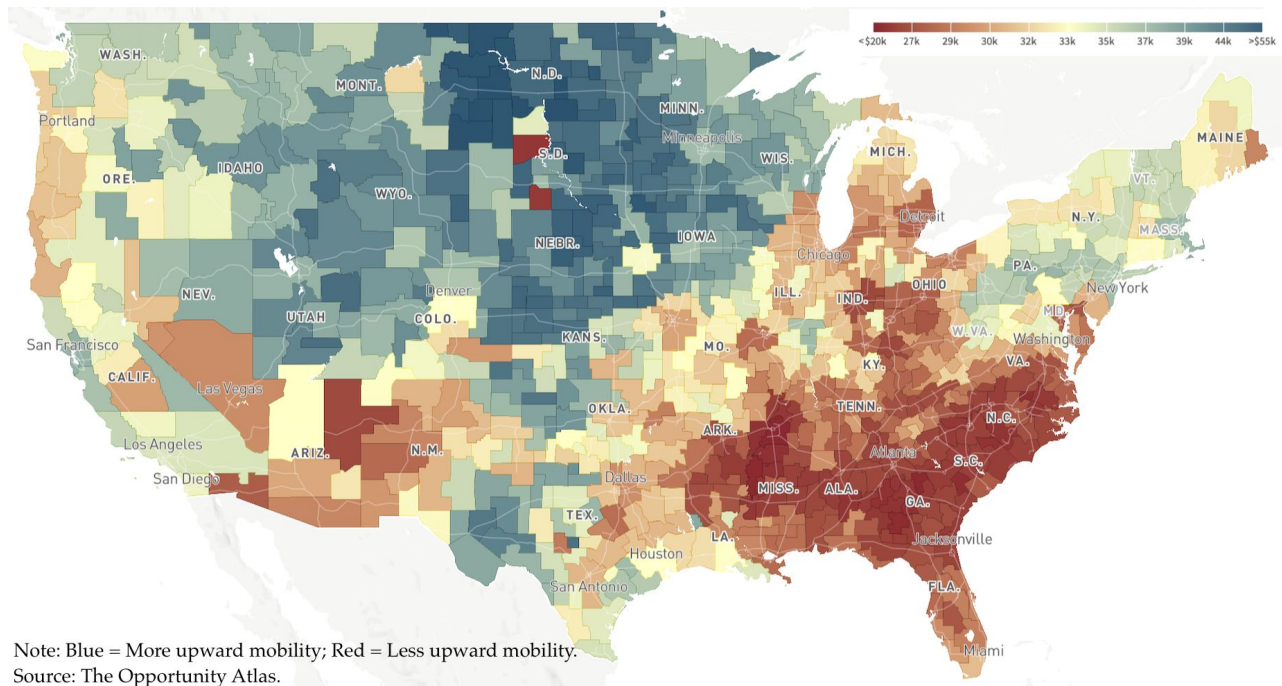
Source: Office for National Statistics (2018)

Within countries: London



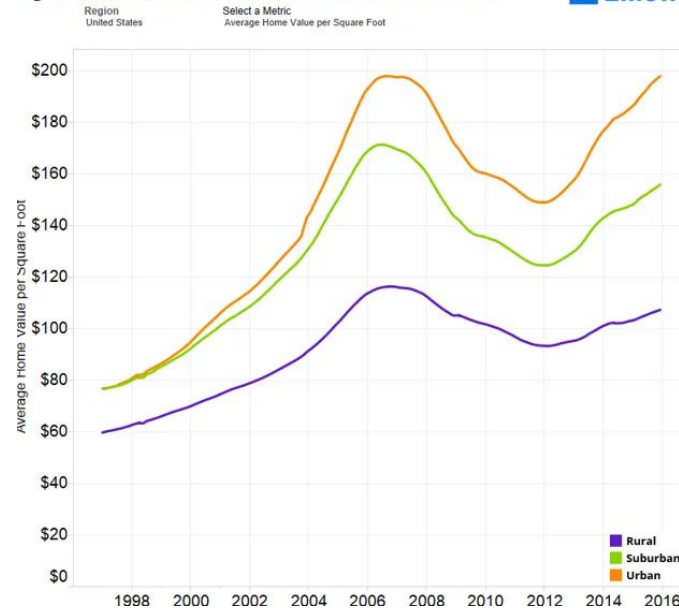
Geography of upward mobility: US

Household income at Age 35 for Children of Low Income Parents

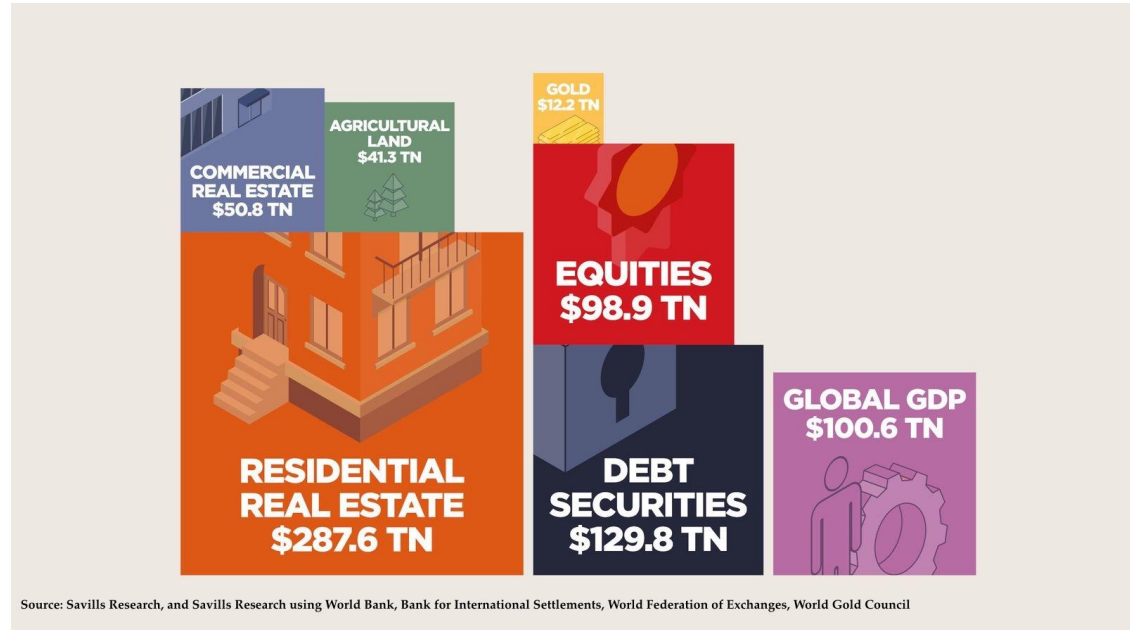


Expensive property located in cities

Figure 1: Urban, Suburban & Rural Home Values Over Time



Property remains the world's biggest store of wealth



Why do we need urban economics?

- Use urban economics to understand
 - Spatial variation in demand/supply of space
 - At regional, urban, local level
 - How the interplay of demand and supply determines
 - Prices and quantities on real estate markets
- Related to a host of practical questions
 - How to maximize welfare?
 - e.g., correct for market failures, internalize externalities
 - How to maximize profits?
 - e.g., is it worth investing/developing at given prices?
- Relevant for policy-makers, planners, investors, developers, etc.

Examples of questions we may cover

(Spoiler alert)

- Why does economic activity concentrates in cities and regions?
- Are big cities good place to work but bad places to live?
- Are cities too big, too small, or just right?
- Why do prices vary within cities?
- How much space do firms and households consume in cities?
- How does spatial structure vary across cities?
- How did the spatial structure of cities change over time?
- Why do firms cluster?
- What determines the vertical dimension of cities?
- Which places are gentrifying and why?

We need urban economics to answer!

Example: Edge cities

- Edge cities are large developments by single developers
 - Satellite cities in major metro areas built on empty land
 - (More on this in Topic 6)
- How to decide where to build an edge city to maximize profits?
 - Land is cheaper further away from center/CBD (good)
 - But: Less connected to firms in the CBD (bad)
 - What about competition between edge and core city?

We need urban economics to answer!

Example: Edge cities

- We need theory to understand
- ... The benefits of being close to the core
 - Urbanization and localization economies
 - Input sharing, labor market pooling, knowledge spillovers
- ... Why rents decline with distance to the center
 - Firms and residents trade access against rents (spatial equilibrium)
- Need good empirical estimates of strength and spatial scope of agglomeration economies and CBD price gradient

We need urban economics to answer!

Example: Building heights

- High-rise buildings involve massive investments
 - Construction costs can be hundreds of millions
 - Land acquisition costs can also be high
- How do we know how tall to build to maximize profits?
 - Taller building (more floor space) generates more rental revenues
 - But: More expensive to build
- How much can we bid for land at given profit margin?
 - Need to know the profit-maximizing building height
 - Then, can think about NPV of investment and how much to bid
 - (More on Seminar 7)

We need urban economics to answer!

Example: Building heights

- We need theory to understand how locational factors determine the market rent in spatial equilibrium
- Need good estimates of
 - (Hedonic) implicit prices of location factors
 - Convexity of construction costs
- Can compute NPV and make an informed bid for land

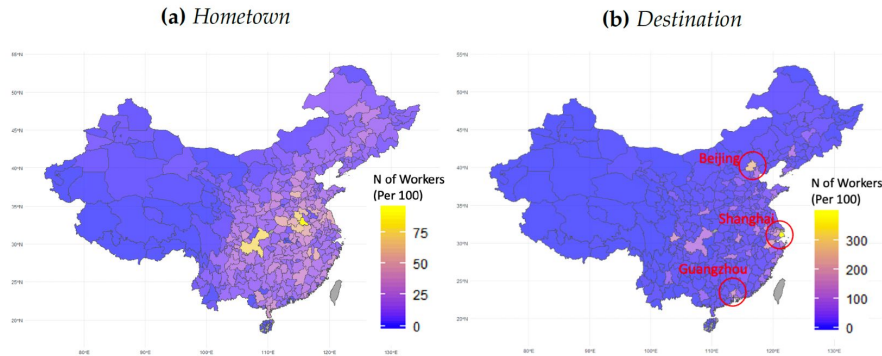
Outline

1. Teaching team and course logistics
2. Motivation: Why urban economics?
3. Case study: Tang (2024)

Case study: Tang (2024)

- *“To Follow the Crowd? Benefits and Costs of Migrant Networks”*
 - Observation: Often migrant workers cluster in the same sectors and locations with peers from the same regions
 - Research question: (*Why?*) What are the benefits and cost of such clustering?
 - Setting: Migrant workers in the gig economy working for a large food delivery platform in China (similar to *UberEats*)
 - Very rich data on millions of migrant workers and their routes

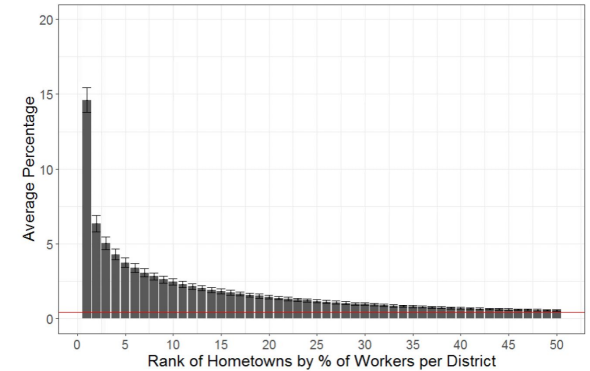
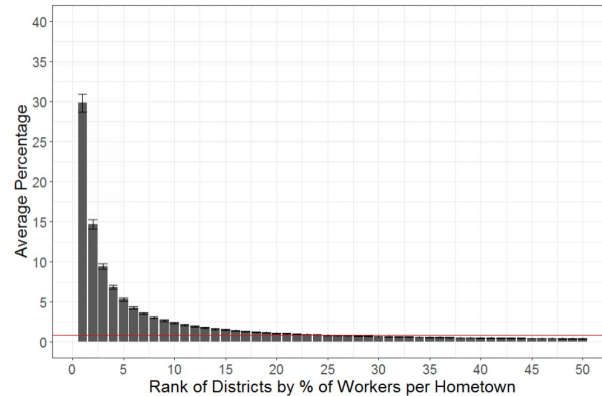
Geographic distribution of food delivery workers



Note: This Figure shows the geographic distribution of delivery workers at the city level in China. I include all active workers between 2020 and 2021 in five large cities. Details of the sample selection can be found in Appendix C.2.1. Figure (a) classifies workers based on their county of birth, and Figure (b) classifies workers based on their current active working cities. Color from blue to yellow represents the increase in the number of workers.

Average clustering level by sending and receiving locations

Figure 1: *Average Clustering Level by Sending Hometowns* **Figure 2:** *Average Clustering Level by Receiving District*

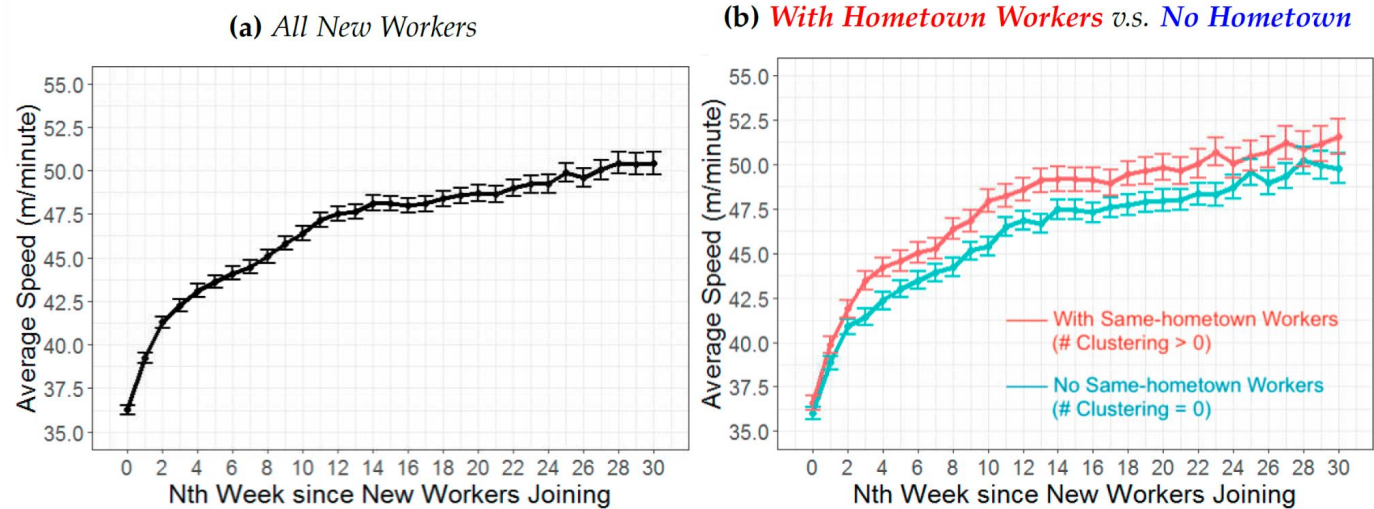


Impact of clustering on learning and productivity

- In food delivery sector: faster delivery speed = more deliveries per hour >>> higher wages
 - Good proxy for worker productivity
- Findings:
 - 1. New workers become more productive over time
 - 2. New workers who cluster with hometown peers become more productive faster than workers not clustering with hometown peers

Impact of clustering on learning and productivity

Figure 3: Productivity Trend of New Workers

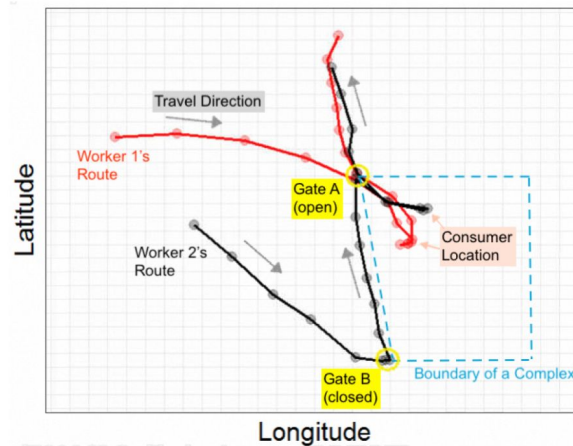


Impact of clustering on learning and productivity

	Restaurant Search Time (min) (1)	Road Driving Speed (min/km) (2)	Consumer Search Time (min) (3)
Own Visit to Restaurant	-1.43*** (0.09)	-0.08 (0.16)	-0.12 (0.10)
Referrer's Visit to Restaurant	-0.51*** (0.14)	-0.06 (0.15)	-0.10 (0.09)
Own Visit to Consumer	-0.21 (0.15)	-0.15 (0.10)	-1.25*** (0.05)
Referrer's Visit to Consumer	-0.10 (0.16)	0.03 (0.12)	-0.58*** (0.09)
Date X Hour X Worker FE	Y	Y	Y
Location FE	Y	Y	Y
BH Control	Y	Y	Y
Ave. Dep. Var.	6.22	10.86	5.93
Observations	5,837,304	5,837,304	5,837,304
R ²	0.43	0.39	0.44

Example of learning from peers

(a) Examples of Going to the Wrong Entrance



(b) % of Workers Going to Wrong Entrances

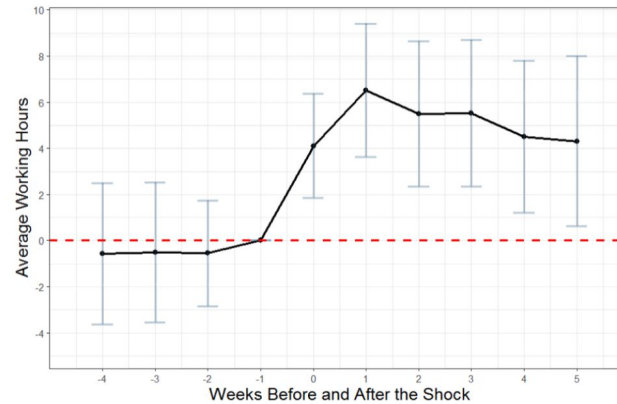


Impact of clustering on income risks

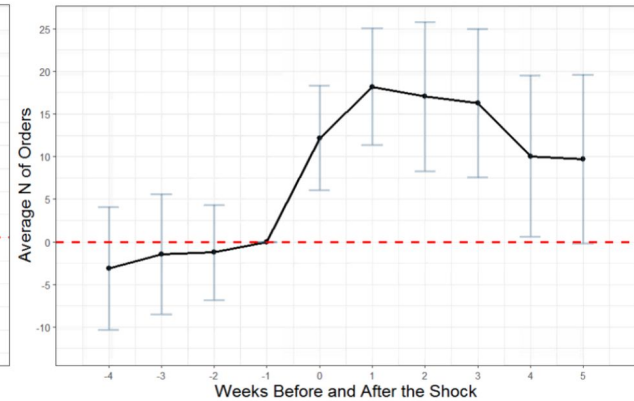
- Migrant workers tend to work longer and function as insurance against adverse shocks in rural areas
 - Send remittances to left-behind family members in hometown
- *Useful context*: Delivery workers are the main source of income for their households + Can quickly adapt labor supply in response to hometown shocks
 - Two types of shocks are analyzed: (i) floods, (ii) lockdowns

Impact of clustering on income risks: Hometown floods

(a) *Labor Supply*



(b) *Earnings (Number of Orders)*



Impact of clustering on income risks

Table 5: *District-level Response to Aggregate Shocks*

Dep. Var.: Weekly Outcomes at District Level				
	Number of Workers	Number of Orders	Total Working Hours	Per-delivery Commission Fee
1{Predicted Shock Share > 15%}	4.925 (3.259)	95.720 (104.525)	148.502*** (33.291)	-0.226 (0.309)
Week FE	Y	Y	Y	Y
District FE	Y	Y	Y	Y
Ave. Dep. Var	132.59	7,233.97	2,408.26	6.41
Observations	14,391	14,391	14,391	14,391
R ²	0.884	0.886	0.886	0.903

Impact of clustering on income risks

$$Y_{ihdt} = \beta \text{ Shock}_{ht} + \delta \text{ AggShock}_{dt} + \alpha \text{ Shock}_{ht} \times \text{ AggShock}_{dt} + \gamma_i + \gamma_{pt} + \gamma_d + \epsilon_{ihdt}$$

Table 6: Congestion: Impacts of Clustering on Delivery Workers' Real Wages during Shocks

Dep. Var.: Average Weekly Outcomes			
	1{ Active }	Ave. Deliveries	Ave. Hours
1{ Hometown Shock}	0.072*** (0.014)	19.939*** (3.912)	6.112*** (1.108)
1{District Shock Share > 15%}	-0.003 (0.022)	-4.870 (2.997)	-0.648 (0.455)
Interaction Term	-0.009 (0.010)	-8.997*** (2.080)	-0.302 (0.378)
Province × Week FE	Y	Y	Y
Worker FE	Y	Y	Y
District FE	Y	Y	Y
Ave. Dep. Var.	0.76	78.10	22.08
Observations	638,810	638,810	638,810
R ²	0.542	0.573	0.582

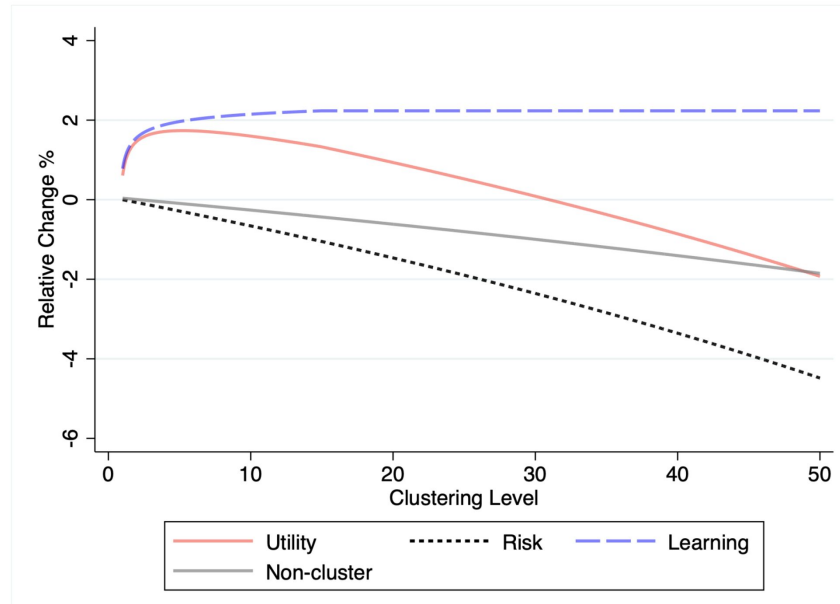
Back of the envelope calculation: During adverse hometown shocks, clustered workers real wages are approximately 10% lower than in normal times

Learning vs. congestion effects?

- Model to quantify trade-off between benefits and costs of migrant networks
- Workers decisions: district, labor supply, and amount of remittances
- Model shows that for a risk-averse worker:
 - Productivity effect $>$ congestion effect, under low clustering
 - Productivity effect $<$ congestion effect, under high clustering
 - Exact magnitudes governed by the risk aversion coefficient

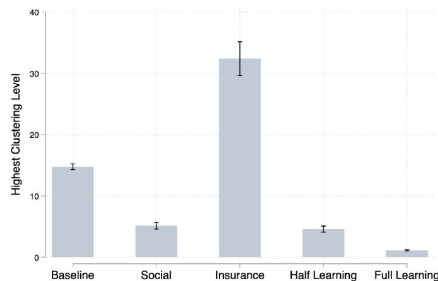
Learning vs. congestion effects?

Figure 9: *Utility as a Function of Clustering*

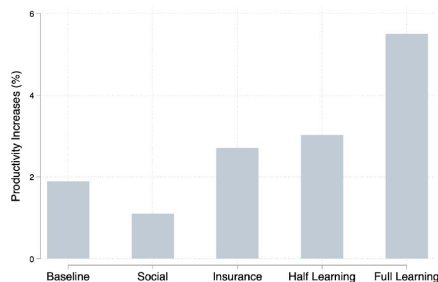


Learning vs. congestion effects?

(a) Maximum Clustering Level



(c) Average Productivity



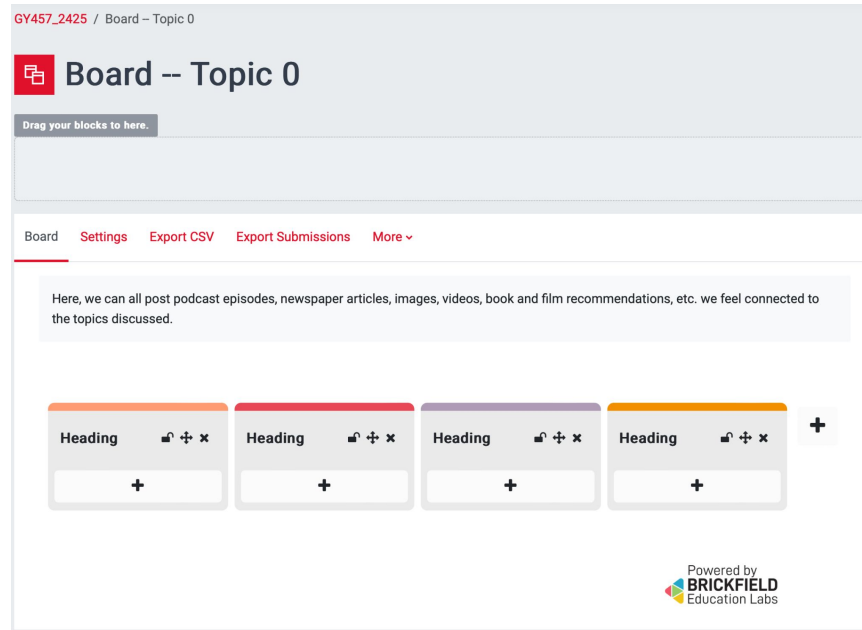
- Social = accounting for learning and congestion externalities
- Insurance = providing migrant workers with insurance to mitigate hometown shocks
- Half learning = providing technology via platform which substitutes half of network-based learning
- Full learning = providing technology via platform which substitutes 100% of network-based learning

Next lecture

- We will go after two broad and fundamental questions:
 1. Why do large industrial industrial regions exist?
 2. Why do cities exist?

Thank you!

Idea: Platform to share content connected to what we are discussing?



- Template of Moodle Board (offline at the moment)
- ... What do you think?

Readings and references

- Optional reading:
 - Glaeser E. L. (2012), The Triumph of the City, Introduction
- References:
 - Tang, Yulu (2024). To follow the crowd? Benefits and Costs of Migrant Networks. Mimeo.